COLLEGE NAME : DONBOSCOCLEGE (CO-ED), YELAGIRI HLLS

COLLEGE CODE : 358

DEPARTENT : BCA

TITLE : Courier Management System (CMS)

TEAM MEMBERS NAME :

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ABSTRACT

Courier Management System (CMS)

Courier and logistics companies need efficient systems to manage shipments, track deliveries, and ensure packages reach customers on time. However, many courier services still use manual methods or separate systems to manage shipment booking, delivery tracking, and billing. These traditional methods can cause delays, data errors, and poor visibility for customers, which reduces overall efficiency. It also becomes difficult for companies to monitor deliveries and maintain accurate records.

The **Courier Management System (CMS)** is a cloud-based application developed using **Salesforce** to solve these problems. The system helps manage all courier operations in one place, including shipment booking, package tracking, delivery agent assignment, status updates, and billing management. This centralized system improves organization and makes data easy to access. It also helps employees work more efficiently by reducing manual tasks.

The application uses custom objects such as **Shipment, Customer, Delivery Agent, and Delivery Status** to store and manage courier information. Automated workflows help update shipment status, assign delivery agents, and manage billing automatically, reducing manual work and improving accuracy. This improves coordination between different departments in the courier service.

The system also provides **reports and dashboards** that help managers track delivery performance and monitor shipment progress. These reports allow companies to analyze delivery efficiency and identify areas that need improvement. Managers can quickly access important data and make better operational decisions. By converting manual processes into an automated cloud system, the Courier Management System improves operational efficiency. It also enhances customer satisfaction by providing accurate and timely delivery information.

PROBLEM STATEMENT

Courier and logistics companies handle a large number of shipments every day and must ensure that parcels are delivered safely and on time. Efficient shipment tracking, delivery management, and communication with customers are very important for maintaining service quality. However, many courier companies still depend on manual processes or disconnected systems to manage shipment booking, parcel tracking, and delivery operations. These traditional methods create several operational challenges and reduce overall efficiency.

One of the major problems faced by courier companies is the lack of a centralized system for managing courier operations. Branch managers often find it difficult to organize shipment data, track delivery progress, and assign delivery agents properly. Because the information is stored in different places or recorded manually, it becomes difficult to maintain accurate and updated records.

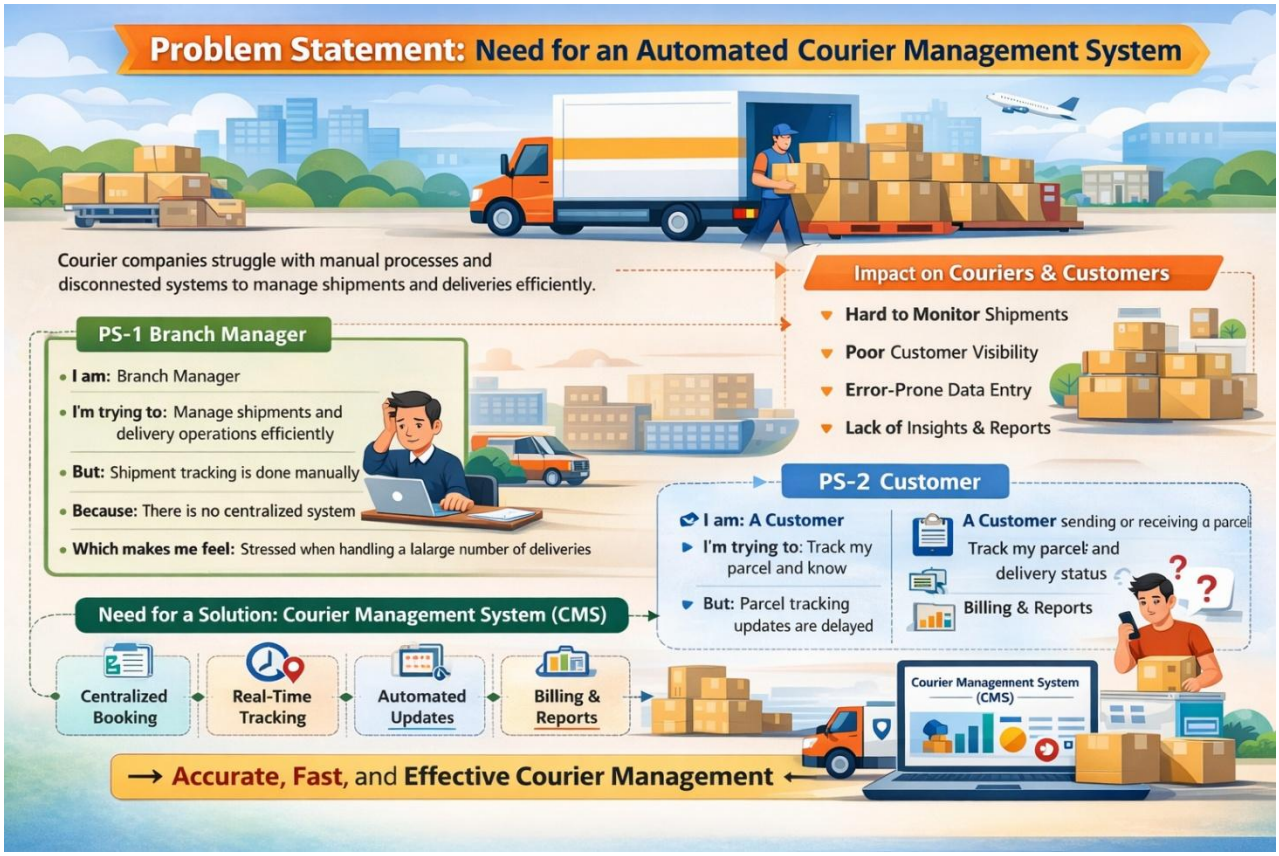
- **I am:** Branch Manager
- **I'm trying to:** Manage shipments and delivery operations efficiently
- **But:** Shipment tracking is done manually
- **Because:** There is no centralized system to manage shipment information
- **Which makes me feel:** Stressed when handling a large number of deliveries

Customers also experience problems when trying to track their parcels. Since shipment updates are entered manually or stored in different systems, customers often receive delayed or incomplete tracking information. This lack of transparency reduces customer confidence in the courier service and makes it difficult for them to know the exact delivery status of their packages.

- **I am:** A Customer sending or receiving a parcel
- **I'm trying to:** Track my parcel and know the delivery status
- **But:** Parcel tracking updates are delayed
- **Because:** Courier systems are disconnected and not automated
- **Which makes me feel:** Uncertain and frustrated about my delivery

Another major issue is the absence of automation and analytical tools in courier management. Manual data entry increases the chances of errors and makes it difficult to generate reports or analyze delivery performance. Managers cannot easily monitor operational efficiency or identify problems in the delivery process.

Therefore, there is a need for a reliable and automated system that can manage courier operations efficiently. The **Courier Management System (CMS)** provides a centralized cloud-based platform built on Salesforce that helps manage shipment booking, parcel tracking, delivery agent assignment, automated status updates, billing, and reporting. This system improves operational efficiency, enhances customer satisfaction, and enables better decision-making through accurate and real-time information.



PROPOSED SOLUTION

To overcome the identified challenges, the proposed solution is the development of a **Courier Management System (CMS)**, a cloud-based courier and logistics management application developed using **Salesforce**. The system is designed to provide a **structured, scalable, and automated platform** for managing courier operations such as shipment booking, parcel tracking, delivery agent assignment, billing, and reporting. By centralizing courier data in one system, the application improves operational efficiency, ensures accurate data management, and provides real-time visibility of shipment status across multiple branches.

Customer Object

This object stores personal and contact details of customers who send or receive parcels. It helps maintain proper customer records and track shipment history.

The Customer object stores details such as:

• Customer Name
• Email Address
• Phone Number
• Address
• Customer Type (Regular / Business)
• City
• State
• Pincode
• Active Status

Shipment Object

This object manages information related to courier shipments and parcel details. It helps track the movement and delivery status of each shipment.

The Shipment object includes details such as:

• Tracking Number
• Customer
• Source Address

• Destination Address
• Package Weight
• Shipment Price
• Shipment Status (Booked, In Transit, Out for Delivery, Delivered, Cancelled)
• Expected Delivery Date
• Assigned Delivery Agent
• Shipment Type (Domestic / International)

Delivery Update Object

This object records the real-time status updates for each shipment during the delivery process.

The Delivery Update object captures details such as:

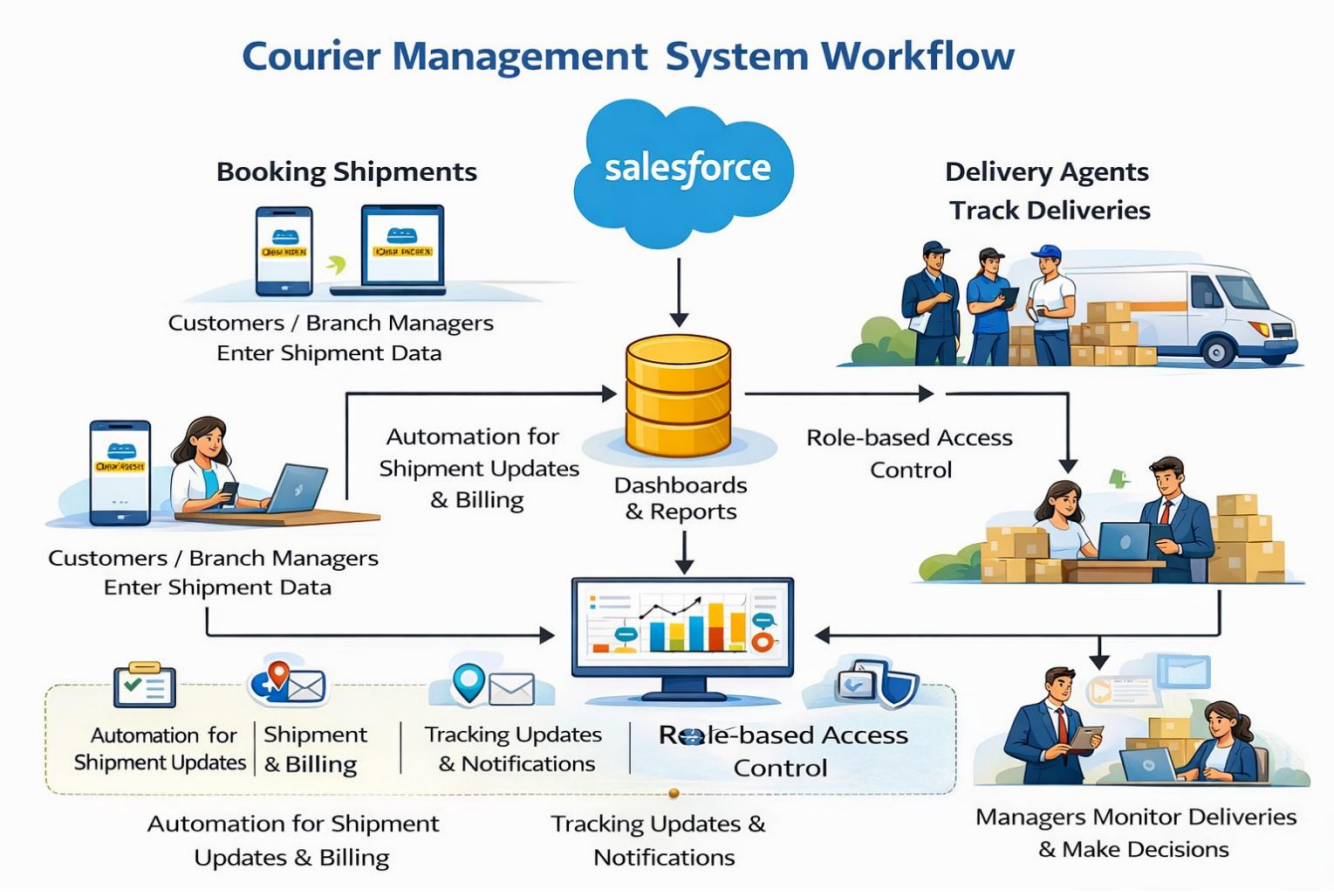
• Update Number
• Shipment
• Status (In Transit, Out for Delivery, Delivered, Delayed)
• Update Date and Time
• Remarks

A **Master-Detail relationship** is established between **Shipment (Master)** and **Delivery Update (Detail)**. This ensures strong data integrity and allows the system to maintain proper tracking records for every shipment. If a shipment record is deleted, all related delivery updates are automatically removed, maintaining consistency in the system.

The application leverages **Salesforce's reporting and analytics engine** to generate real-time insights such as shipment status summaries, agent performance reports, and branch-wise revenue analysis.

Thus, the **Courier Management System (CMS)** transforms manual courier management processes into a **structured, automated, and intelligent logistics management system**, improving operational efficiency, delivery tracking accuracy, and customer satisfaction.

WORK FLOW DIAGRAM



SUMMARY

The **Courier Management System (CMS)** is a cloud-based application developed using Salesforce to improve courier and logistics operations. It helps organizations manage shipment booking, parcel tracking, delivery agent assignment, and billing in a centralized system. By replacing manual processes with automation, the system improves operational efficiency and reduces errors. It also ensures better coordination between different branches and departments.

The system uses several custom Salesforce objects such as Customer, Shipment, Delivery Agent, Branch, Delivery Update, and Invoice to manage courier data effectively. These objects store important information related to customers, parcels, delivery agents, and billing. Relationships between these objects ensure proper data organization and consistency. This structured data model helps manage courier operations smoothly.

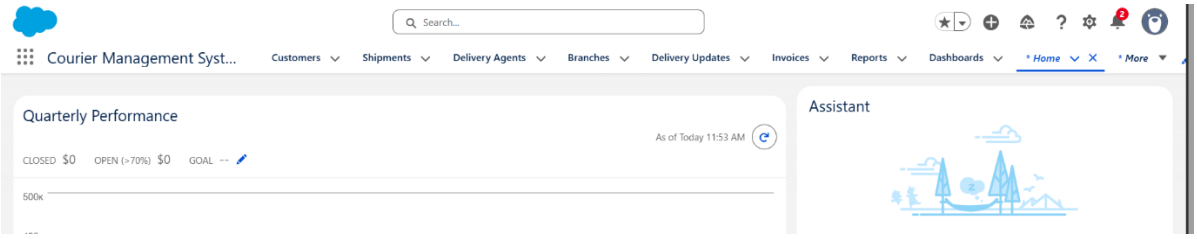
Automation features such as flows, validation rules, and automated status updates help reduce manual work and ensure accurate data entry. Real-time shipment tracking allows customers and managers to monitor parcel movement and delivery progress. The system also provides automated billing and notifications, improving communication and service reliability.

Additionally, reports and dashboards provide valuable insights into shipment status, delivery performance, and branch revenue. Managers can analyze operational data and make better business decisions. These visual tools help track courier activities and identify areas that require improvement.

Overall, the Courier Management System transforms traditional courier management into a modern digital platform. It improves transparency, enhances operational efficiency, and increases customer satisfaction through real-time tracking and automation. By using Salesforce technology, the system supports scalable courier operations and helps organizations manage deliveries more effectively.

SCREENSHOTS

HOME PAGE:



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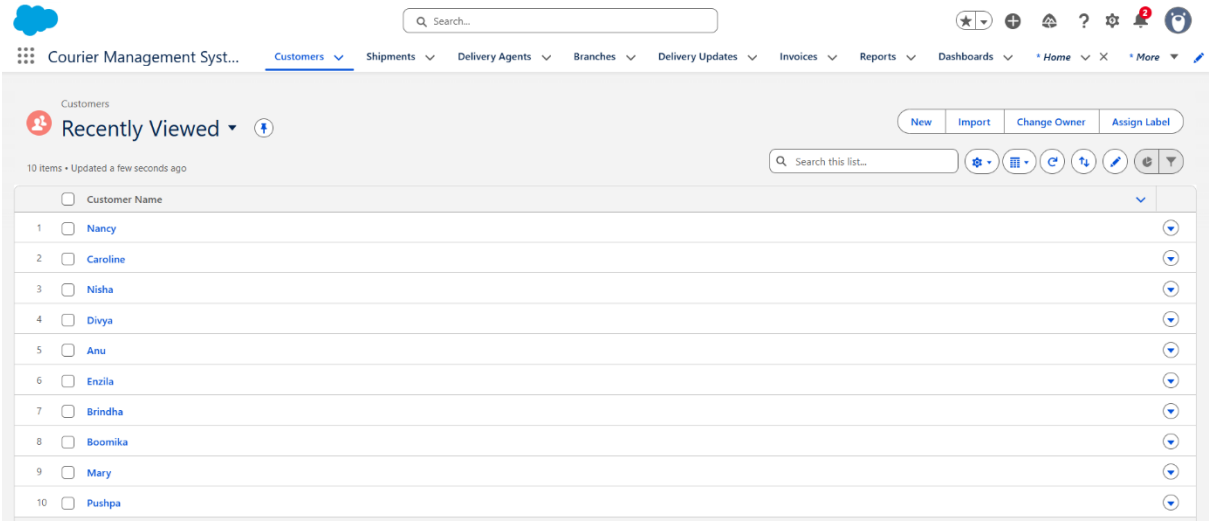
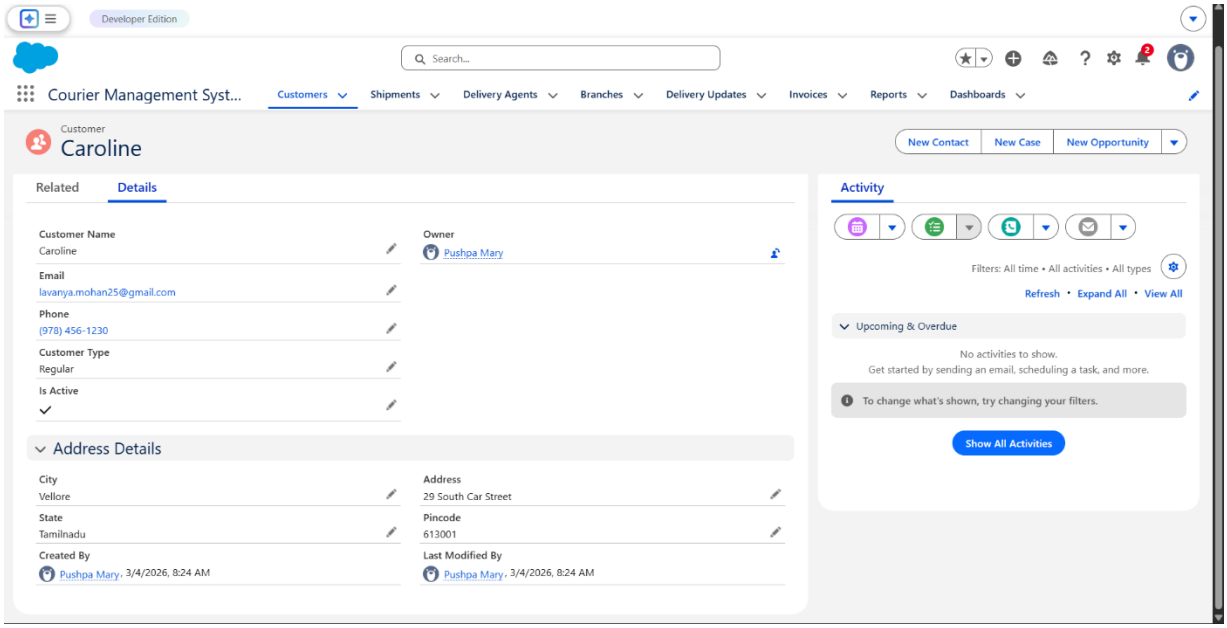
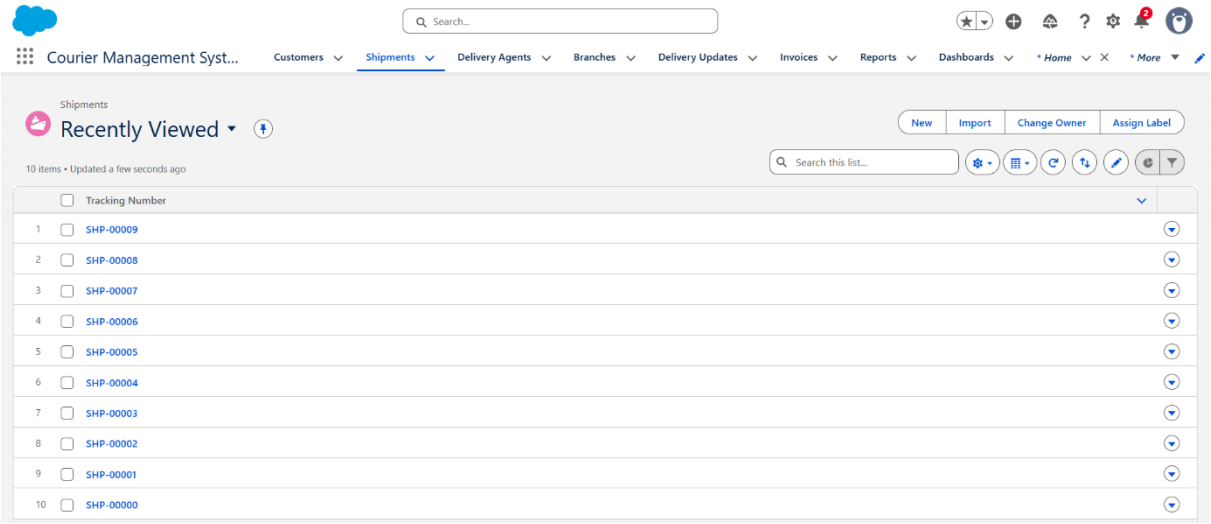




SHIPMENTS :


Courier Management Syst...
Customers
Shipments
Delivery Agents
Branches
Delivery Updates
Invoices
Reports
Dashboards
Home
More
Shipments
Recently Viewed
10 items • Updated a few seconds ago
New
Import
Change Owner
Assign Label
Search this list...
Tracking Number
SHP-00009
SHP-00008
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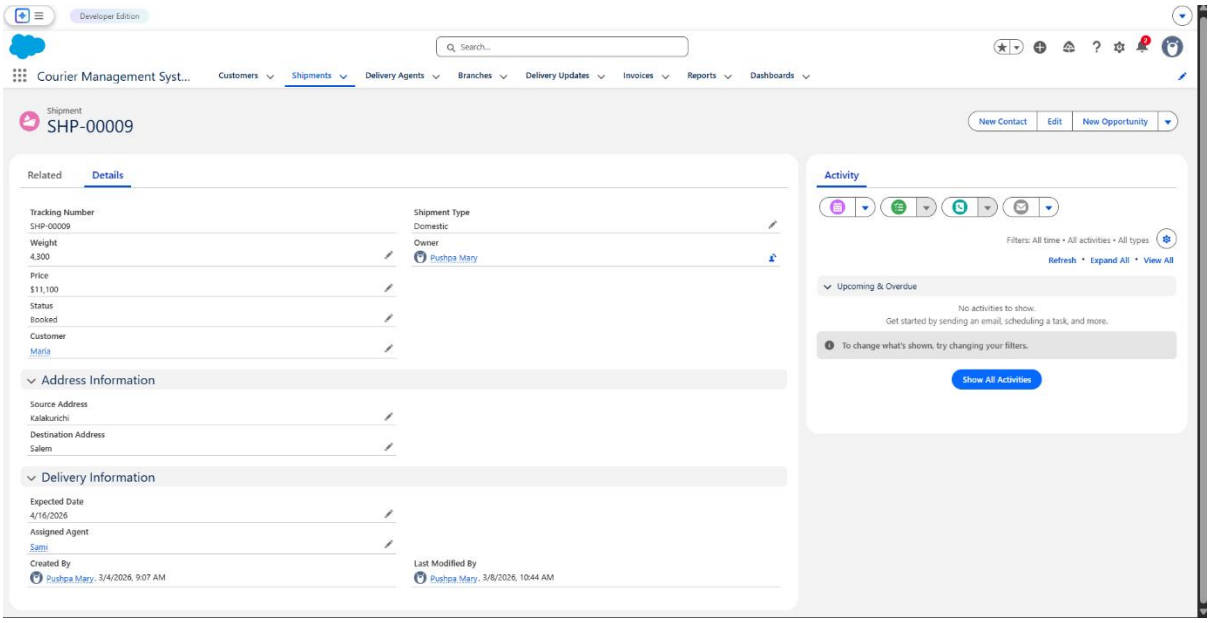
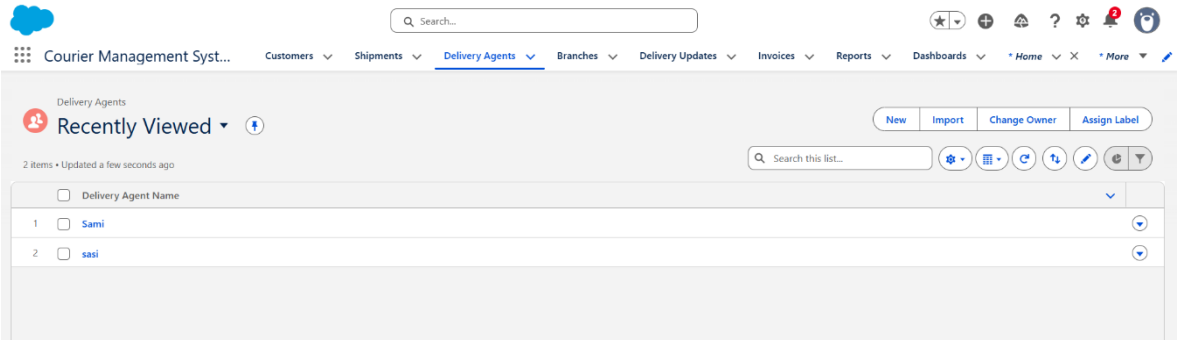
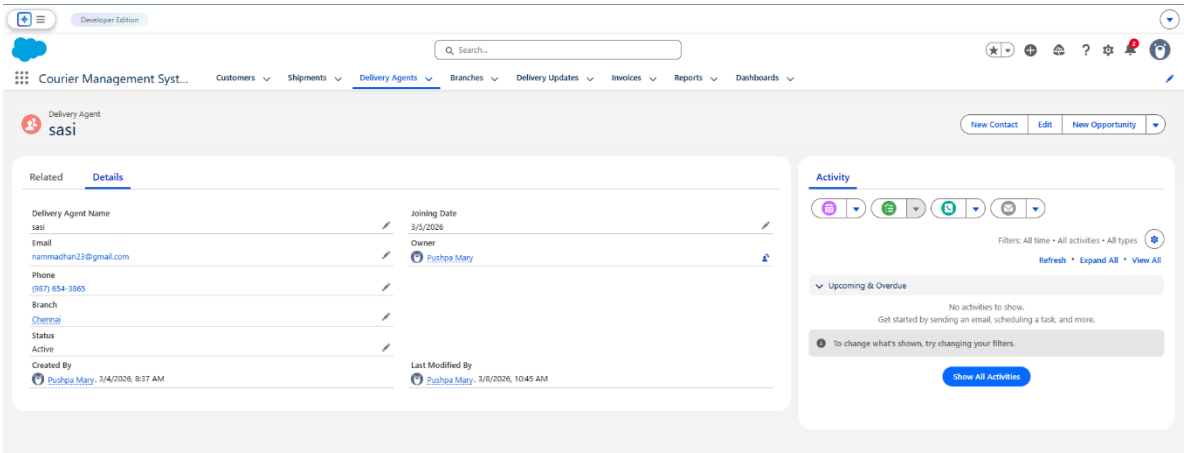
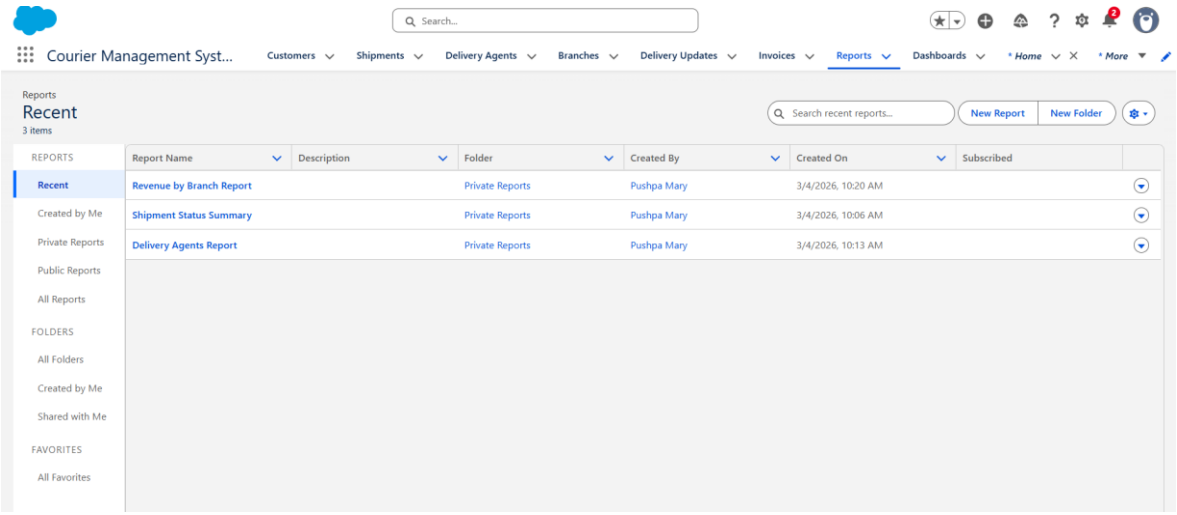

Developer Edition
Courier Management Syst...
Customers
Shipments
Delivery Agents
Branches
Delivery Updates
Invoices
Reports
Dashboards
Shipment
SHP-00009
New Contact
Edit
New Opportunity
Related
Details
Tracking Number
SHP-00009
Shipment Type
Domestic
Weight
4,300
Owner
Pushpa Mary
Price
$11,100
Status
Booked
Customer
Maria
Address Information
Source Address
Kalakurichi
Destination Address
Salem
Delivery Information
Expected Date
4/16/2026
Assigned Agent
Sami
Created By
Pushpa Mary, 3/4/2026, 9:07 AM
Last Modified By
Pushpa Mary, 3/8/2026, 10:44 AM
Activity
Filters: All time • All activities • All types
Refresh • Expand All • View All
Upcoming & Overdue
No activities to show.
Get started by sending an email, scheduling a task, and more.
To change what's shown, try changing your filters.
Show All Activities

DELIVERY AGENTS :

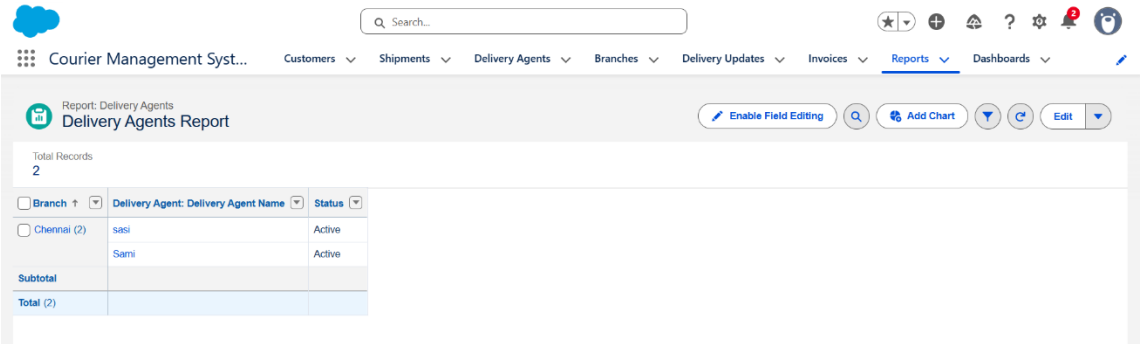




REPORTS PAGE :

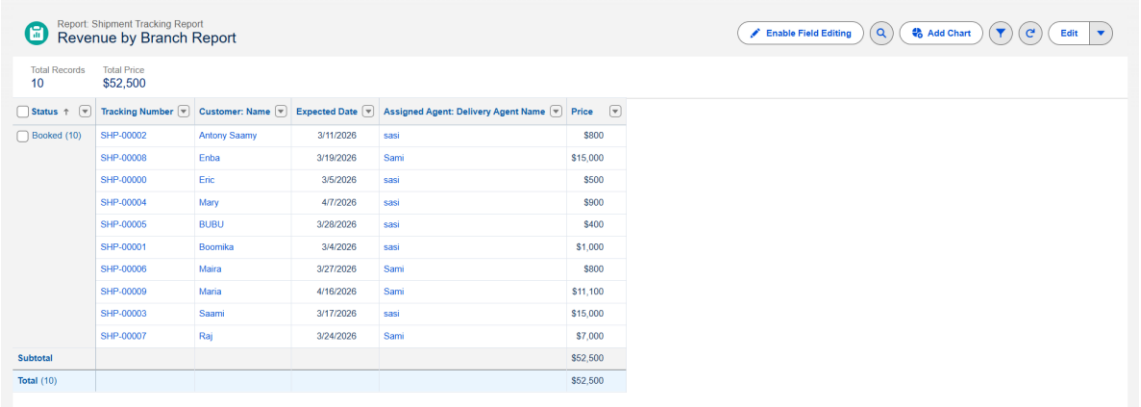


DELIVERY AGENT REPORTS :



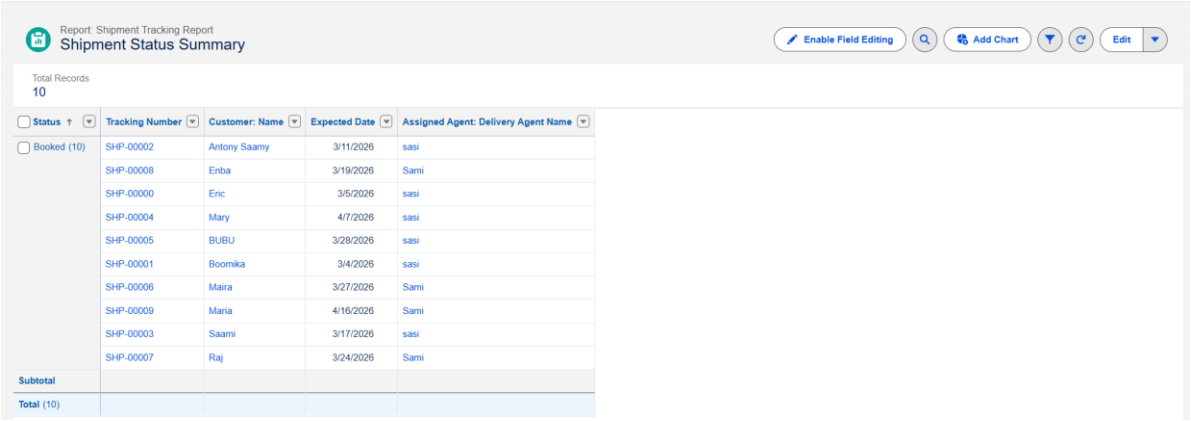
Branch	Delivery Agent: Delivery Agent Name	Status
Chennai (2)	sasi	Active
	Sami	Active
Subtotal		
Total (2)		

REVENUE by BRANCH REPORTS :



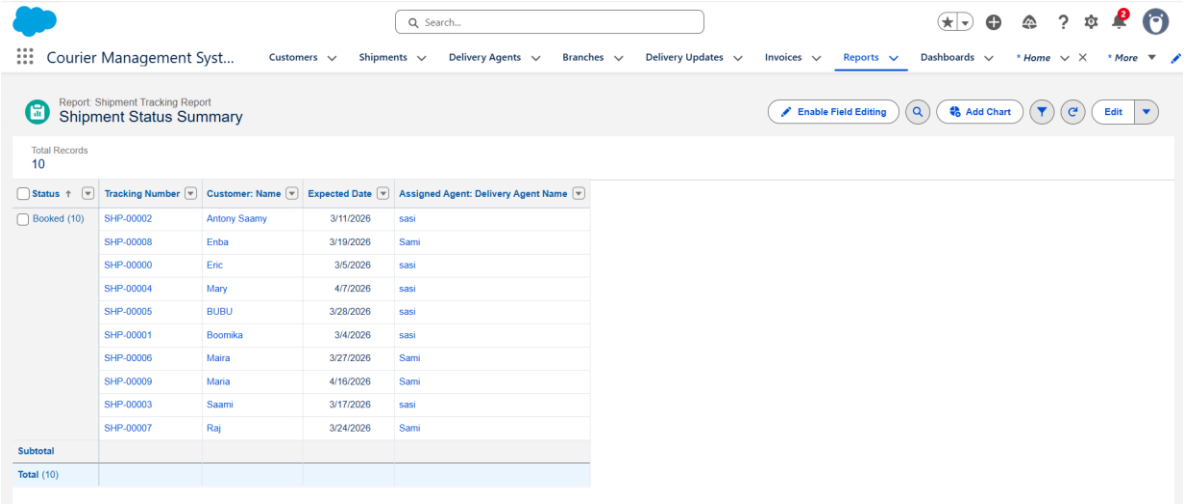
Status	Tracking Number	Customer: Name	Expected Date	Assigned Agent: Delivery Agent Name	Price
Booked (10)	SHP-00002	Antony Saamy	3/11/2026	sasi	$800
	SHP-00008	Enba	3/19/2026	Sami	$15,000
	SHP-00000	Eric	3/5/2026	sasi	$500
	SHP-00004	Mary	4/7/2026	sasi	$900
	SHP-00005	BUBU	3/28/2026	sasi	$400
	SHP-00001	Boomika	3/4/2026	sasi	$1,000
	SHP-00006	Maira	3/27/2026	Sami	$800
	SHP-00009	Maria	4/16/2026	Sami	$11,100
	SHP-00003	Saami	3/17/2026	sasi	$15,000
	SHP-00007	Raj	3/24/2026	Sami	$7,000
Subtotal					$52,500
Total (10)					$52,500

SHIPMENT STATUS SUMMARY :

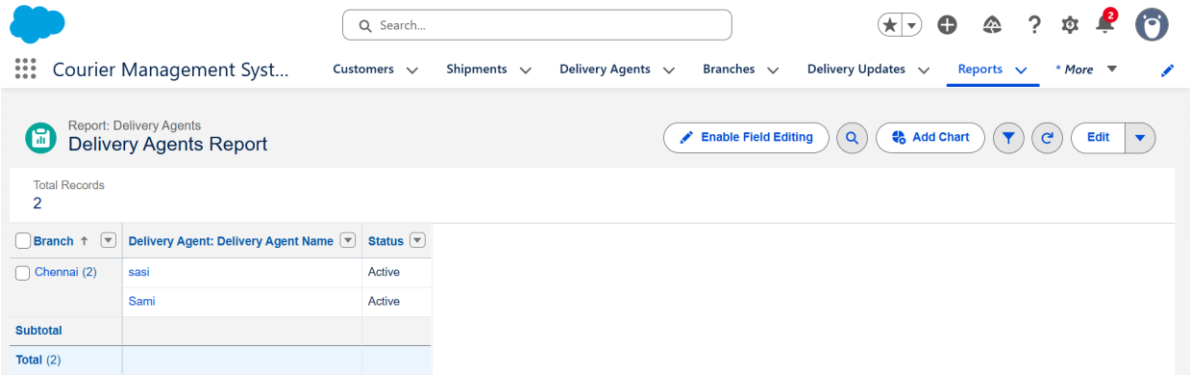


Status	Tracking Number	Customer: Name	Expected Date	Assigned Agent: Delivery Agent Name
Booked (10)	SHP-00002	Antony Saamy	3/11/2026	sasi
	SHP-00008	Enba	3/19/2026	Sami
	SHP-00000	Eric	3/5/2026	sasi
	SHP-00004	Mary	4/7/2026	sasi
	SHP-00005	BUBU	3/28/2026	sasi
	SHP-00001	Boomika	3/4/2026	sasi
	SHP-00006	Maira	3/27/2026	Sami
	SHP-00009	Maria	4/16/2026	Sami
	SHP-00003	Saami	3/17/2026	sasi
	SHP-00007	Raj	3/24/2026	Sami
Subtotal				
Total (10)				

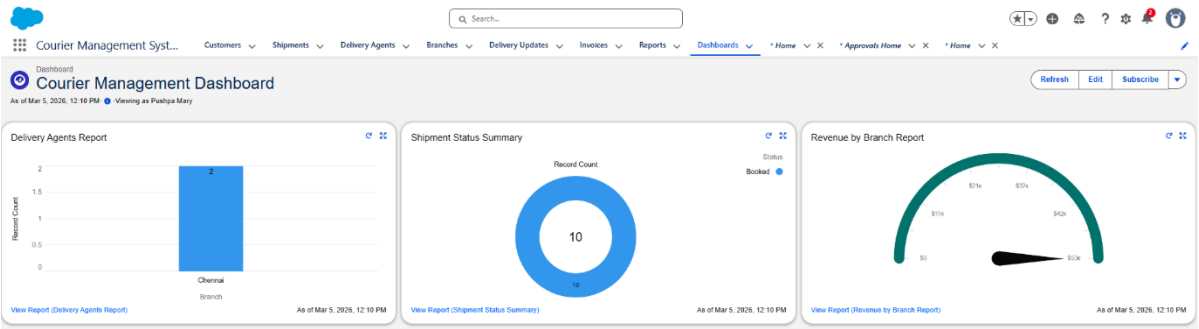
SHIPMENT STATUS REPORTS :



DELIVERY AGENT REPORTS :



DASHBOARD PAGE :



CONCLUSION

The **Courier Management System (CMS)** is designed to improve the efficiency of courier and logistics operations by using the Salesforce platform. It replaces traditional manual methods with an automated and centralized system. This helps organizations manage shipments, customers, and delivery agents more effectively.

The system provides features such as **shipment booking, delivery tracking, agent assignment, and billing management**. These functions help courier companies monitor parcel movement and delivery progress in real time. As a result, operational delays and manual errors are reduced.

By using **custom objects, relationships, and automation tools**, the system ensures that courier data is organized and updated accurately. Validation rules and automated workflows help maintain data quality and reduce manual effort. This improves the reliability of courier operations.

The **reports and dashboards** generated by the system provide useful insights into shipment status, delivery performance, and revenue. Managers can analyze this information to monitor business performance and make better decisions. These analytics help improve overall operational efficiency.

Overall, the Courier Management System successfully transforms manual courier processes into a **structured and automated digital system**. It enhances operational efficiency, improves customer satisfaction, and supports better management of courier services. The system demonstrates how Salesforce technology can be used to optimize logistics operations.